Question	Answer	Marks
2(a)	e.g. period = 3 / 2.5	**C1**
	frequency = 0.83 Hz	**A1**
2(b)	light (damping)	**B1**
2(c)	at 2.7 s, A_0 = 1.5 (cm)	**B1**
	energy = $\frac{1}{2} m \times 4\pi^2 f^2 A_0^2$	**B1**
	$= \frac{1}{2} \times 0.18 \times 4\pi^2 \times 0.83^2 \times (1.5 \times 10^{-2})^2$ $= 5.51 \times 10^{-4}$ (J)	**C1**
	at 7.5 s, A_0 = 0.75 (cm)	**B1**
	energy = $\frac{1}{4} \times 5.51 \times 10^{-4}$ **or** energy = $\frac{1}{2} \times 0.18 \times 4\pi^2 \times 0.83^2 \times (0.75 \times 10^{-2})^2$	**C1**
	energy = 1.38×10^{-4} (J) change = $(5.51 \times 10^{-4} - 1.38 \times 10^{-4}) = 4.13$ J	**A1**

Question	Answer	Marks
3(a)(i)	signal consists of (a series of) 1s and 0s or offs and ons, or highs and lows	B1